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Interactive Augmented Reality to Extend Students' Vector Experience

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First-year engineering students show difficulties in connecting basic vector algebra with manipulations using the graphical representation. Experiencing vectors in real applications can improve the process of learning vectors. Thus, we introduce a unique approach for using digital technologies, namely augmented reality. We utilized a video see-through head-mounted display to develop a vector manipulation tool enabling users to select and place virtual vectors in the real environment. The aim was to see how students react when experiencing vectors in augmented reality. After using the prototype, students answered a questionnaire about this experience. Results show that students experiences are positive and they are motivated to use the prototype in the course. Further, we investigated opinions about vector concepts from students in the trial group and other students attending the same course in vector algebra. The results indicate some differences in what students spontaneously associate with the concept vectors. Moreover, this study provides insights into how such investigations can be elaborated in the future.

1. INTRODUCTION

Dropout rates are especially high in degree programs where mathematics courses are taught in the first year of study (Heublein, 2014). As reported in Saadati and Celis (2023), mathematics is a challenging subject for most students in technical studies and the universities are doing great efforts towards developing mathematics knowledge and skills, mainly by influencing students' motivations. Bingolbali et al. (2007) found that engineering students see mathematics as a tool and wish to see how mathematics is applied. Pepin et al. (2021) argue that integrating real-life problems may help engineering students to see the use of mathematics. Timcenko & Thomsen (2023) describe how, using virtual reality (VR), one can expose students to tangible representations of abstract mathematical concepts in support of both students' motivation and their retaining of mathematical knowledge.

A specific mathematical challenge is that undergraduate engineering students show misunderstandings and false assumptions when learning vector algebra. This problem is occurring in undergraduate engineering studies at Karlstad University. Students' conceptual understanding of vector concepts tends to be low, according to Taqwa and Rahim (2022), who argue that vector learning should be focused on the meaning of vectors in various representations. Further, we see the importance of motivation to enhance learning and the use of emerging technologies, such as VR, as means to attract and motivate students (Su et al., 2022)

According to Doerner et al. (2022), a visionary way of describing the "perfect virtual reality (VR)" is simulation that cannot be distinguished from reality. While creating such a simulation is not yet technologically possible, he argues that the more technological definition of VR from Bryson (1993) is still applicable: "[…] the use of three-dimensional displays and interaction devices to explore real-time computer-generated environments". In other words, an illusion is created so that users feel like they are present in another world. This is usually achieved by head-mounted displays (HMD), which are displays the user wears directly in front of the eyes (Broll et al., 2022). The difference between VR and augmented reality (AR) is the level of immersion: in VR, the user is completely immersed in a virtual world, thus she cannot see the real world in her surroundings. On the other hand, in AR there are digital objects superimposed upon the real world which is still visible by the user (Azuma, 1997). While the real-time interaction of AR is similar to VR, reality is not replaced but combined with virtuality, creating a coherent whole (Doerner et al., 2022). The AR approach presented in this study is interactive by having virtual content anchored at a real-world position, so the user can move spatially to change the perspective on the digital content. The interaction design of the AR prototype of this study allows the user 1. to walk closer to a virtual vector placed in the room to take a better look at it; 2. to create vectors using hand controllers; 3. to perceive the distance between real objects in the room using vectors. Finally, an AR HMD allows the user to interact with peers in the room while viewing superimposed digital objects (see Figure 1).

In a comparative study, Demitriadou et al. (2020) found that adding VR or AR to a course in primary education increases interactivity and student motivation in learning mathematics. Both technologies have potential in mathematics education. Moreover, Bulut & Borromeo Ferri (2023) conducted a systematic literature review on AR in mathematics. They discussed that all recent studies report multiple benefits of AR systems for education, which goes from increased student performance to benefits for personal development of the student, such as teamwork and creativity (Bulut & Borromeo Ferri, 2023; Garzón & Acevedo, 2019; da Silva et al., 2019). They stress that instructors must be supported implementing AR, free AR resources and authoring tools are needed, and physical and virtual learning should be connected (Bulut & Borromeo Ferri, 2023).

In their work, Schutera & Schnierle et al. (2021) present a prototype study focussing on vector geometry in the 3D space. Students can scan a QR code on exercise sheets to visualise the related problem through a mobile AR application. Compared to our study, students interact with digital content through a touch screen interface inside of the application, while we use direct hand interaction. The authors conclude that the students' understanding of the 3D space was strengthened and the application was not stressful to use for students. Tan & Tay (2022) tested a desktop AR system for teaching vectors in an online class during the COVID-19 pandemic. By utilising the webcam of the user's laptop, virtual planes and vectors were added to real physical content (e.g., when holding a book in front of the webcam, a virtual plane parallel to the book was presented, plus vectors in the direction of the book). In conclusion, adaption and feedback from both teachers and students were positive.

1.1 Research objective

Inspired by the work of Timcenko and Thomsen (2023), this project investigates whether experiencing vector algebra in head-mounted display (HMD) augmented reality (AR) can increase students' motivation of learning vector algebra. We use AR for displaying vector algebra, since it allows to interact with superimposed digital vectors in the real space using hand controllers. Therefore, the research question is:

What are the student's experiences when interacting with Vectors in an Augmented Reality environment?

1.2 Paper outline

Section 2 describes our hands-on trial method, prototype used, and data collection. This is followed by Section 3, which presents the results of a post-trial questionnaire and the results of a follow-up survey. Section 4 discusses these results together with limitations and how to address these in future iterations. Finally, we conclude the study in Section 5.

2. METHOD

This explorative study targeted first-year engineering students at an on-going university course that partly contained vector algebra. The study had two parts: an AR prototype hands-on trial including a post-trial questionnaire, and a follow-up vector survey at the end of the course. The prototype trial aimed at the students' experience of AR and was conducted before the students learned about vectors in the course, while the follow-up survey aimed at students' beliefs about vector concepts after learning vector algebra in the course, comparing responses from students that took part in the prototype trial and other students of the mathematics course who did not take part in the trial. We refer to these students as a "control group", although we have not measured a baseline for each group before the trial, as described in the classical experimental design (Clark et al., 2021).

In the prototype trial study, students were exposed to an AR prototype with the featured ability to let users place an object in the room and extend a chain of vectors from it (Figure 1 & 2). The sum of the vectors could be calculated (Figure 3). We argue that this equipment provides a "realistic" demonstration of vectors' directions and extensions because these appear in mid-air exactly where the user places them, even if the user walks around in the room, and because these "vectors" are directly manipulated via the hand controllers which appears to the users as two human hands. After using the prototype, the students answered a questionnaire about this experience. Later, we investigated beliefs about vector concepts from some students in the treatment group and some other students attending the same course in vector algebra in order to see if this first trial left any indication of differences in students' perception of vectors. Open-ended questions in our data collection were analysed using thematic analysis according to Clark et al. (2021), where data were gradually put into codes and later generalised into themes, which we present in Section 3.2 & 3.4.

2.1 Prototype hands-on trial study

Students of the mathematics course were split into exercise groups of around 30. One group was randomly selected to be invited to take part in the hands-on trial. Sub-groups of maximum six students could select from four dates on a date-picker website. After the first study day it was clear that there would be few attendances in the following dates. Therefore, students from another exercise group were invited to the study. Two subgroups consisted of six students, two subgroups consisted of four students, and the last two subgroups consisted of only one student each. Eventually the six subgroups with a total of 22 students volunteered to participate in the trials.

Figure 1: User drawing a vector in front of a mirror in the laboratory. The green sphere is grabbed by the hand controller and moved rightwards by the user, changing the vector's direction.

Prototype. The functionality of the prototype is to create a path of vectors using your hands in AR. Vectors are placed virtually in the physical surroundings of the user. For this, the user selects the starting position of the first vector by grabbing and placing a red sphere using a hand controller. Then, a green sphere can be moved which automatically spans a vector between the red and green sphere. This vector is locked as soon as the green sphere is dropped by the user. Moving the green sphere again spans a new vector that starts at the ending point of the prior vector (Figure 2). Repeating this behaviour creates a path of n vectors. By pressing a

button on the hand controller, the user can toggle the sum vector that spans between the starting point of the first vector and the ending point of the last vector. Also, vector directions are presented under each vector. For the sum vector, its length is also presented, which can be utilised to measure the distance between objects in the user's physical surroundings. The user can remove all vectors and start again.

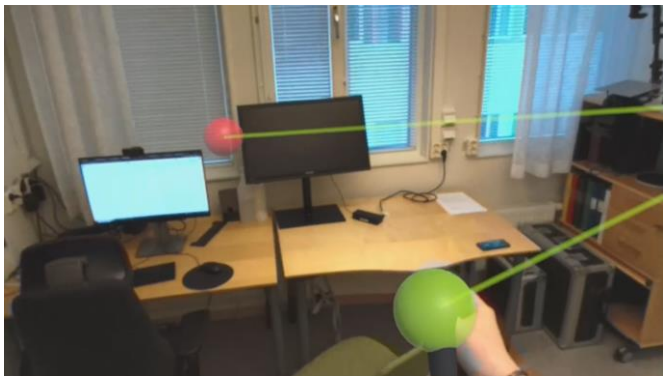

Figure 2. Cutout of a user's POV drawing the second vector. The background shows the laboratory where the study took place.

Setup. The trial study took place in the Usability Laboratory of the Information Systems department at Karlstad University. HMD and its two hand controllers were placed on a table next to an ordinary computer monitor that showed the view of the student using the HMD to the rest of participants of the session. This image was recorded together with sound emitted by the students. At the beginning, the instructor set up the broadcasting from the HMD to the monitor. Then, the first student of the group was introduced to the HMD by the instructor, followed by introducing this student to the functionality and controls of the prototype (that is the vector manipulation application), including the sum vector and the precision vector values. Then the student was free in what to do. Finally, the student was instructed to explain the controls when switching the HMD to the next student. The instructor will step back and observe while only interfering with the students if asked or necessary for technical assistance.

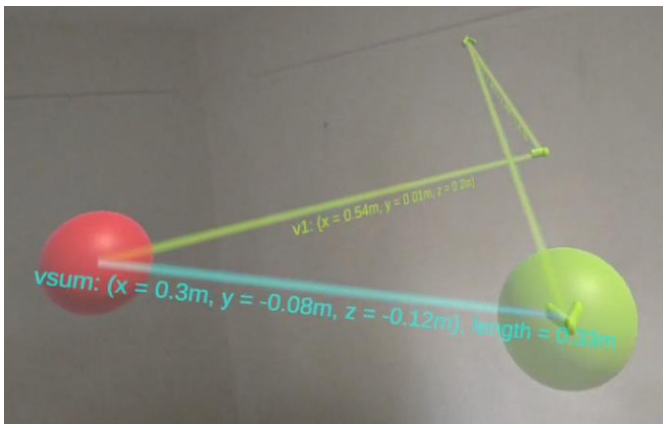

Figure 3. Cutout of a user's POV toggling the sum of vectors and positional information. The blue vector is the sum of the three green vectors previously drawn by the user.

Technical setup. In this trial, students use the Meta Quest Pro HMD (Meta, 2023) and two hand controllers. The HMD utilises video see through (VST) technology. According to Broll et al. (2022), VST devices create an experience similar to VR, although, "a video image of reality is inserted in such a way that the user has the impression that she can look at the world around her through glasses". To broadcast the HMD's view to the monitor, we used the in-built broadcasting functionality from Meta. Moreover, we used the open-source software tool 'OBS Studio' to record the screen image of the monitor and the sound of the students in the room. The post-trial questionnaire was created using the Karlstad University web-based survey tool. Participants could use three computers to answer the post-trial questionnaire.

Post-trial questionnaire. After students have completed the AR experience, they answered a post-trial questionnaire containing both open-ended and closed-ended questions. The questions were used to gain feedback from students about their experiences. The questions were designed together with the vector course teachers. See Table 1 for close-ended questions and Table 2 for open-ended questions. In summary, questions aimed on *prior experience with AR/VR*, *adoption of AR technologies* in general (TC.1) and for learning activities (TC.2, TC.3), and *perception of the specific AR vector experience*. These contain usefulness (TC.4, TC.6); learning outcomes (TC.7, TO.3); positive (TC.5, TO.1), negative (TO.2) and surprising (TC.8, TO.4) impressions; and possible improvements (TO.5). Participants could give final thoughts at the end of the questionnaire (TO.6). TC stands for **T**rial-**C**losed-ended question, and TO for **T**rial **O**pen-ended questions.

2.2 Follow-up vector survey

The second part of the study took place a few weeks after the hands-on trial and at the end of the vector algebra course. An online follow-up survey containing open and closed questions was open to any students volunteering to answer, that is, both students who took part in the hands-on trial and those who did not, to check if there is any indication of differences in students' perception of vectors. See Table 3 & 4 for a list of questions asked. Questions focused on students' *factual knowledge of vectors* (FC.1, FC.2, FC.3, FO.1, FO.2), perceived *usefulness of vectors* (FC.4, FC.5), their *learning self-efficacy and motivation* (FC.7, FC.10), and their *perceptions of the vector algebra course* (FC.6, FC.8, FC.9, FO.3). One question (FC.11) aims on how students visualise vectors while doing calculations. Finally, participants that took part in the trial study were also asked if AR should be embedded in the course in the future (FC.12). FC stands for **F**ollow-up **C**losed-ended questions, and FO for **F**ollow-up **O**pen-ended questions.

2.3 Ethical considerations

This trial study was granted the ethical approval by Karlstad University ethical advisors (HNT 2023/612). Participation in the trial as well as responding to the surveys (post-trial and follow-up) was voluntary. Students participating in the trial were informed of the study and had to give their consent to the data collection and recordings

(image and sound) during the trial. These recordings were used to capture the students' behavioural responses to the prototype. Apart from that, no personal data were collected. Furthermore, the course's teacher was not present, so students were not pressured to participate and neither should they feel they were being assessed.

3 RESULTS

In the following, we present results of the data collection. Results are grouped into closed-ended (Section 3.1) and open-ended responses of the post-trial questionnaire (Section 3.2), as well as closed-ended (Section 3.3) and open-ended responses of the follow-up vector survey (Section 3.4).

3.1 Post-trial questionnaire – closed-ended responses

Table 1 presents the treatment group responses to Likert-based closed-ended questions we asked participants in the post-trial questionnaire. The following paragraphs present the highlight results of their responses. The following subsections present the highlights of the closed-ended responses.

Prior experience with AR/VR. 16 out of 22 participants indicated that they have never used AR or VR in a scholarly setting before. However, 15 out of 22 participants reported prior experience using AR/VR in a personal setting compared to scholarly usage.

Adoption of AR technology. Participants show consensus (22 out of 22 participants agree) that they want to use AR technology again, both in general (TC.1) and in a course setting (TC.2). Most participants (21 out of 22) want to use AR technology more often to learn in the future. (TC.3).

Perception of AR vector experience. Also, the majority of students (21 out of 22) would recommend the AR vector experience to other students in the course to a certain degree (TC.4) and perceive it as useful (TC.6). About two thirds of students (15 out of 22) disagreed that they learned something about vectors, while about one third (7 out of 22) agreed (TC.7). Students showed strong consensus (22 out of 22) in perceiving the experience to be fun (TC.5). No consensus (10 students disagreeing, 12 students agreeing) was found in being surprised about the way they experienced vectors (TC.8).

3.2 Post-trial questionnaire – open-ended responses

Of the 22 students participating in the trials, several responded to the six open-ended questions (found in Table 2) that we asked after the AR vector experience (22 responses to TO.1, 20 responses to TO.2, 12 responses to TO.3, nine responses to TO.4, 19 responses to TO.5, and 13 responses to TO.6). We identified three themes in responses to questions TO.1, TO.3, TO.4 and TO.6 which correspond to positive feedback, namely *visualisation opportunities*, *rich interaction experience* and *motivation*. Additionally, a theme of *future improvements* in responses to questions TO.2 and TO.5 were identified.

Visualisation opportunities. When asked about what participants liked about the prototype (TO.1), they most dominantly argued for how vectors are visualised in the prototype: The 3-dimensionality utilising the stereoscopic view enabled through head-mounted displays was mentioned 8 times as a factor of enjoyment (response to TO.1) and 2 times when asked for factors of surprise (response to TO.4).

Visualisation offering new perspective was one point highlighted by a participant who stated: "What was most interesting to me was how you got a new perspective on the vectors as a whole. For someone who has a hard time getting a picture in their head for mathematical problems, this can be a good tool for them. I therefore believe that this tool can be a good tool for teaching maths" (TO.1).

Experiencing something new due to the visualizations offered by the prototype was also a factor contributing to the positive feedback, as 3 participants appreciated that they experienced something for the first time (TO.1). 2 students were surprised by the AR experience since it was new to them (TO.4).

Additionally, students mentioned opportunities created by the nature of AR. A total of 3 participants stated that they enjoyed altering the real environment with virtual objects (TO.1), which could be seen as an informal description of AR technology. For example, 1 participant enjoyed "[b]eing able to see reality while being able to see vectors in space".

The precise positional tracking of the HMD and hand controllers used in the prototype study also contributed to positive feedback. It allowed participants to utilise vectors for measuring objects or distances. This functionality (utility) was appreciated by 2 participants (TO.1). "What I appreciated most was that we got to test it ourselves and try out how it worked and that the measuring tools were so precise" – a proband.

The prototype enabling learning was present in responses. 6 participants stated that the AR experience led or could lead to understanding of vectors (3 responses each to both TO.1 and TO.4).

Rich interaction experience. Participants can interact with vectors in the AR prototype, which, compared to dealing with vectors on paper, offered a richer experience and therefore a more hands-on interaction. The experience is rich because students can discuss with peers, use their hands, move around using their body, draw vectors in three dimensions, etc. While 2 participants indicated in their responses that they praise the interaction in general (TO.1, TO.4), 3 participants highlighted the way vectors can (nearly) be touched (2 responses TO.1, 1 response TO.4). The freedom of the user being able to move spatially to change their view on the drawn vectors using their body is mentioned 2 times (TO.1). For example, a participant stated that "[i]t was interesting to be able to walk around and see the vectors 'still in the air'". Another student stated that "[i]t not only feels like, but you can actually touch [the vectors]".

There was a total of 5 responses that praised the practical approach of the AR experience to learning vectors (TO.1). In other words, participants enjoyed that they could try-out the prototype themselves or could get "hands-on" experience. For example, a participant described that they enjoyed most to get "hands on experience with otherwise plain and boring lines on a paper".

Motivation. 2 participants described an impact on their motivation to learn about vectors after experiencing them in the AR prototype (TO.1): For example, "[t]he technology made it seem more fun to learn about vectors".

Furthermore, 1 student described the prototype to motivate learning by overcoming fears (TO.4): "Seeing it in this 'format' or this point of view makes them less 'intimidating'."

Another factor of motivation was the description of having had fun in during the experience. When asked to give final thoughts (TO.6), 8 people responded that the experience was "cool", "fun", or "interesting", of which 5 students also showed interest to try it out again in possibly new contexts.

Future improvements. This section summarises students' comments on future improvements of the AR experience which stem from responses to TO.2 and TO.5 (Table 2).

The first group of improvements tackles limitations of the setup. A total of 4 students stated that they dislike how long it takes to get used to being immersed in the AR experience. In addition, there were visual problems for many students using the prototype. This was mentioned 9 times that a student saw their environment with a blur or flickering which was disliked (TO.2). Moreover, 3 students mentioned that this should be improved (TO.5). There were problems with adjusting the HMD to fit the participants' heads; a wrong fit can lead to blur. Secondly, it was not possible for people with eye impairment to wear glasses under the HMD device. Finally, the described flickering is reasoned in a technicality of the Quest pro device: the camera-AR system creates flickering.

Another improvement factor is described by visibility problems of information caused by the design of the prototype. Stated by 10 participants (4 responses TO.2, 6 responses TO.5), this mainly aims on the vector information text placed under vectors in the experience: this text is sometimes mirrored to the user. 4 students suggested that the text should always be directed towards the user to increase readability. One of these students proposed a way of getting the information of all vectors on an "info pop-up".

There were 4 participants mentioning that they were missing a task structure during the AR experience to facilitate learning (2 responses TO.2, 3 responses TO.5; one student mentioned this in responses to both questions). For example, one participant was curious about how the tool is expected to be used (TO.5).

We identified certain new prototype features that would enhance the experience in participants' responses: Firstly, the interaction could be improved so that users have more freedom (2 responses TO.2, 1 response TO.5). For example, using both hands and changing vectors after they have been drawn. Secondly, there are three improvements as new mathematical content: a grid / coordinate system (4 responses TO.5); more vector calculations like vector subtraction, multiplication, etc. (2 responses TO.5); and angles, allowing geometrical tasks (1 response TO.5).

3.3 Follow-up vectors survey – closed-ended responses

Tables 3 & 4 present results of students' responses, both trial group and control group. They are grouped into responses from students who took part in the prototype trial study (trial group) and those who did not (control group).

Most students from the trial group (10 out of 11) stated that they visualise vectors in their head during vector calculations, while four out of 14 control group students disagreed to this statement (FC.11). Moreover, all students from the trial group (11 out of 11) would like to see AR technology being part of the vector course in the future (FC.12).

Factual vector knowledge. All students from both groups (11 out of 11, 14 out of 14) agree to certain degrees that they understand what vectors are (FC.1). Participants from the trial group show slightly higher confidence regarding vectors (9 out of 11 state confidence) than the control group (7 out of 14 state confidence) (FC.2). Both groups are leaning towards confidence (10 out of 11 agree from trial group, 10 out of 14 from control group) in working with different representation forms of vectors (FC.3).

Perceived vector usefulness. Participants of the trial and control groups mostly agree (respectively, 8 out of 11, 11 out of 14) to vectors being helpful to students in the future (FC.4). Additionally, both groups mostly agree (10 out of 11 trial group, 13 out of 14 control group) that vector calculations can be helpful to solve engineering problems (FC.5).

Vector learning motivation and self-efficacy. Both groups mostly disagree (10 out of 11 trial group, 13 out of 14 control group) that they cannot understand vectors even with high efforts (FC.7). Responses regarding motivation to learn advanced vector applications in the future show mixed agreement (7 out of 11 trial group, 8 out of 14 control group) (FC.10).

Course evaluation. 12 out of 14 students from the control group agreed that the course was sufficient to learn vectors; responses from the treatment group are mixed (7 out of 11 agree) (FC.6). Similar responses are given regarding if the course was motivating to learn vectors (8 out of 11 agree in trial group, 11 out of 14 in control group) (FC.8). We will not comment on responses to FC.9 since it is ambiguously formed if they are happy because they do not like vectors or happy that they made the first step into vector knowledge.

3.4 Follow-up vectors survey – open-ended responses

Responses to the questions FO.1 and FO.2 (Table 4) can be summarised in certain ways how vectors are described by the participants. These contain *analogies* for vectors and their *characteristics*. We identified certain *learning needs* from responses to question FO.3 (Table 4). Most students gave responses to FO.1 (10 out of 11 trial, 12 out of 14 control) and FO.2 (11 out of 11 trial, 11 out of 14 control), less participants responded to FO.3 (5 out of 11 trial, 9 out of 14 control).

Vector analogies. When describing vectors, students used the term "arrow" to describe a vector in response to FO.1 (7 out of 11 trial, 3 out of 14 control); this term was only used two times in response to FO.2 (1 out of 11 trial, 1 out of 14 control). The term "line" was less present with 3 responses to FO.1 (1 out of 11 trial, 2 out of 14 control).

Vector characteristics. Participants mentioned certain vector characteristics: Most dominantly, they stated that vectors have a *direction*, responding to FO.1 (7 out of 11 trial, 10 out of 14 control) and FO.2 (8 out of 11 trial, 7 out of 14 control). This is followed by vectors having a *magnitude* in responses to FO.1 (8 out of 11 trial, 4 out of 14 control) and FO.2 (2 out of 11 trial, 7 out of 14 control) contained the terms force, power, length, or distance. Statements that vectors describe a *movement* or speed were less dominant, with responses to FO.1 (1 out of 11 trial, 3 out of 14 control) and FO.2 (2 out of 11 trial, 0 out of 14 control). Participants described vectors as *non-stationary*, meaning that they can be moved around, with responses to FO.1 (1 out of 11 trial, 1 out of 14 control) and FO.2 (2 out of 11 trial, 2 out of 14 control). Finally, students (1 out of 11 trial, 1 out of 14 control) responded to FO.2 stating that vectors contain *multiple points*.

Vector learning needs. Students mentioned four vector learning needs not fulfilled by the course in response to FO.3: Firstly, participants from both groups (1 out of 11 trial and 3 out of 14 control) mentioned that measures should focus on the 3-dimensionality of vectors. Secondly, vectors should be visualised in more and different ways (1 out of 11 trial, 2 out of 14 control). Moreover, students asked for digital tools being used in the course (1 out of 11 trial, 2 out of 14 control). Finally, students mentioned that practical relevance of vector problems has been missing (1 out of 11 trial, 1 out of 14 control). Certain participants denied the question (1 out of 11 trial, 4 out of 14 control).

4. DISCUSSION

Overall, there was a consensus in results of the prototype trial study: students were strongly motivated to use AR in the course, and the follow-up vector survey confirmed AR motivation, which aligns with the findings on motivation of Timcenko & Thomson (2023). Similar to Su et al. (2022), our findings show that the use of an emerging technology in the course is appreciated by students. We can see this as an opportunity to integrate the tools further into the course's learning activities. Students pointed to certain improvements of the prototype, see further discussion in section 4.1. The prototype's impact on vector perspectives was less noticeable: the differences between trial and control group were indirect. In the following, we discuss student responses in relation to key findings.

The results of the questions TC.2, TC.4 and FC.12, dealing with students' interest of using AR in scholarly settings, might indicate that trial group students kept their positive beliefs on wanting to use AR in university courses while taking the vector course. Responses to FO.3 show that students have a need for digital systems that visualise 3D. However, this was mentioned less by students who used our prototype. We see an opportunity of improving the understanding of vectors in 3D through AR as Schutera & Schnierle et al. (2021) found a positive impact of AR on understanding the 3D space.

While in the trial study students show interest in embedding AR in university courses in general, they state that it should be deployed in the mathematics course they took part in during this study. However, some responses to question FC.6 show that the trial group were less agreeable to the statement that the vector algebra course gave them the tools they needed to learn vectors than the control group. Although we did not investigate the reason for that, we think that this could be an indication that they would have liked to use AR more in this course. Probably, the experience set their expectations higher of what the course could have offered. Thus, future efforts should be made on embedding AR experiences into university courses. If done correctly, this can address the need for clarifying practical relevance of learning content stated by students in our study. Thus, the way vectors are represented in future iterations of the prototype should follow real-life examples, as supported by Bingolbali et al.'s (2007) and Pepin et al.'s (2021) findings.

Responses to FO.1 and FO.2 showed that trial group students used the term "arrow" as a metaphor for vectors more often than the control group. This is an indicator that the visual form of the vectors in AR might shape the mental model students have about vectors. In a follow-up study, this could be further investigated by adding multiple metaphors of vector representation to the prototype, using it as a controlled variable. For example, vectors could be represented as lasers that are shot, ropes that are pulled, or rockets that fly at a certain speed. Furthermore, responses to FO.1 and FO.2 aligned with the terminology used in the course lectures and course book. Furthermore, participants from the trial group mentioned magnitude without being prompted to and the control group mentioned it when prompted to explain the difference between vectors and points. When including AR into a course curriculum, choosing appropriate metaphors should align with the lectures and other course material.

During the trials, there were instances of students having a moment of realisation or being in awe. For example, during explaining the vector sum to students in one trial session, a student responded by saying "ahhhh" in a tone expressing understanding. The student confirmed that he could connect the explained content to prior knowledge from high school. Next to this, multiple students said "wooow" in a tone

expressing awe/excitement shortly after being exposed to the virtual vectors in AR. One could argue that something like a 'wow' moment wears off. However, 15 out of the 22 students had already AR/VR experience in personal setting. Thus, for the future, it should be investigated whether these are key moments that students will remember and whether it leads to motivation to work more with vectors in this and other formats.

4.1 Limitations and considerations

The results of this study might be biased due to the sampling of participant for the treatment group: since we could not randomly select students from the beginners engineering course to take part in the treatment group study, we advertised the study at the beginning of two classes. This means that participation was based on students' personal interest and motivation – an unmotivated student is less likely to take part in the study in their free time. Thus, the results might not be representative for all students. This could explain that all students were interested in using AR again in the future. Therefore, the control group is not truly a control group as described in the classical experimental design (Clark et al., 2021).

The interest bias might also apply to the results of the follow-up survey that did not show a lot of differences between trial and control group students; again, students were asked to complete the survey after class in their free time. In future iterations, this is fixable by using time during the classroom activity to make students complete the survey. On the other hand, it would be harder to select participants randomly for a one-hour treatment group study trial due to ethical concerns. One way to achieve that would be to embed the study in the course curriculum, for example, as a laboratory task that could be randomly assigned to certain groups of students.

A technical limitation to this study is the use of the video-mediated see-through AR technology. Compared to an optical see-through glass display, the used device (Quest Pro) captures the user's field of view with a camera and displays it on the binocular displays, leading to blurriness of the surroundings for some participants (see results). In future iterations, an optical see-through AR device should be used, which was not available at the university facilities at the time of prototype creation.

Moreover, we should improve the experience to decrease blurriness for participants. While there must be time addressed in each experimental session to adjust the HMD to each participant wearing it, special measures could be needed for visually impaired participants, depending on the model of HMD used in the study. In our case using the Meta Quest Pro, participants should have been instructed how to change the vertical distance between the two binocular displays, and a spacer for people wearing glasses should have been available.

Furthermore, we observed that students were most likely to show boredom during the trials after they have used the HMD. They interacted less with the group, watched the monitor less and sometimes looked at their phones. In contrast, students who were last in wearing the HMD did not show this boredom, probably because they were still interested in how the experience would be in immersion next to only watching it on the screen. This is an interesting lesson for the application in regular teaching. It makes it relevant to question if the group size of 6 is a suitable number for such a

	# Responses (n = 22)					
Closed-ended question	**strongly disagree**	**disagree**	**slightly disagree**	**slightly agree**	**agree**	**strongly agree**
TC.1: I want to use AR technology again.				1	11	10
TC.2: I would like to have more courses where AR technology is used.				1	15	6
TC.3: In the future, I want to use AR technology more often to learn.			1	3	12	6
TC.4: I would recommend the AR experience to other students in this course.			1	4	10	7
TC.5: The AR experience was fun.					10	12
TC.6: The AR experience is useful for students.			1	6	13	2
TC.7: I learned something about vectors today.		5	10	5	2	
TC.8: The way I experienced vectors today was surprising to me.		2	8	5	7	

Table 1: Prototype study closed-ended questions agreement level.

Open-ended question	**# Responses (n = 22)**
TO.1: What did you enjoy the most during the experience?	22
TO.2: What did you dislike the most during the experience?	20
TO.3: What did you learn about vectors today? Give at least one example.	12
TO.4: What was surprising to you? Give at least one example.	9
TO.5: In your opinion, what should be added to the AR experience to improve it?	19
TO.6: Additional thoughts or comments:	13

Table 2: Prototype study open-ended questions.

	Trial group (n = 11)						Control group (n = 14)					
Response: / **Closed-ended question:**	**strongly disagree**	**disagree**	**slightly disagree**	**slightly agree**	**agree**	**strongly agree**	**strongly disagree**	**disagree**	**slightly disagree**	**slightly agree**	**agree**	**strongly agree**
FC.1: I understand what vectors are.				1	8	2				2	8	4
FC.2: I do not feel confident about vectors. (-)	2	1	6	1	1		2	4	1	6		
FC.3: I can work with vectors in different representation forms.		1		9	1			1	2	5	4	1
FC.4: Understanding vectors will help me in the future.		1	1	2	5	1			2	8	2	1
FC.5: Vector calculations can be helpful to solve engineering problems.	1				4	6			1	1	7	5
FC.6: This course gave me everything I needed to learn about vectors.	1	2	1	4	3				2	7	5	
FC.7: No matter how much effort I put in, I cannot understand vectors. (-)	4	5	1		1		6	6	1		1	
FC.8: This course motivated me to learn about vectors.	1		2	6	2			3		8	3	
FC.9: I am happy that we are done with vectors in this course. (-)		2	1	5	3			2	4	2	4	1
FC.10: I am motivated to learn more advanced applications of vectors in future courses.	1	1	2	4	3			2	4	5	3	
FC.11: When doing vector calculations, I visualise the vectors in my head.			1	4	5	1		3	1	4	5	1
FC.12: Augmented Reality should be used in this course in the future. *(Only asked to treatment group)*				5	5	1						

Table 3: Vectors survey closed-ended questions agreement level, responses grouped by prototype study attendance.

Open-ended question	**# Responses trial group (n = 11)**	**# Responses control group (n = 14)**
FO.1: If I were to explain vectors for a 10-year-old, I would say:	10	12
FO.2: If I should explain the difference between a point and a vector, I would say:	11	11
FO.3: Is there anything in the course that you would have liked to see more of when learning about vectors?	5	9

Table 4: Vectors survey open-ended questions.

group experience, even if the cost of equipment would tend to drive universities to this or larger experience groups. Still, as students pointed out in the trial survey, there was no task given in the trial. In future iterations, we will try to embed a problem that has to be solved cooperatively to keep students engaged.

For future studies, we suggest a comparative design between HMD-based AR, mobile AR, and the use of 3D software for teaching a mathematical problem. As HMD-based AR is, to this day, less accessible than mobile AR tools or 3D software like GeoGebra, such a study design could inform teachers under what circumstances higher efforts of implementing HMD-based AR to their teaching is justified.

5. CONCLUSIONS

This study presents promising results for using AR as a means to motivate and enhance the learning experience of undergraduate engineering students. Students enjoyed the AR prototype and wanted to use it again in the mathematics course. As for the learning of vectors, our findings show that AR experiences could motivate learning and that AR exposure of 3D could shape how vectors are perceived.

Based on these results, we continue to develop the prototype in for the mathematics course. Specifically, we want students to use the prototype to solve a practical problem as part of a laboratory assignment. Our results resemble Timcenko and Thomsen (2023) regarding students' motivation. We believe that deploying interactive AR in more courses could have an impact on mitigating the high dropout rates of students.

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